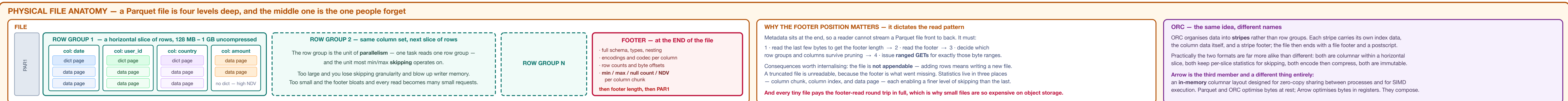


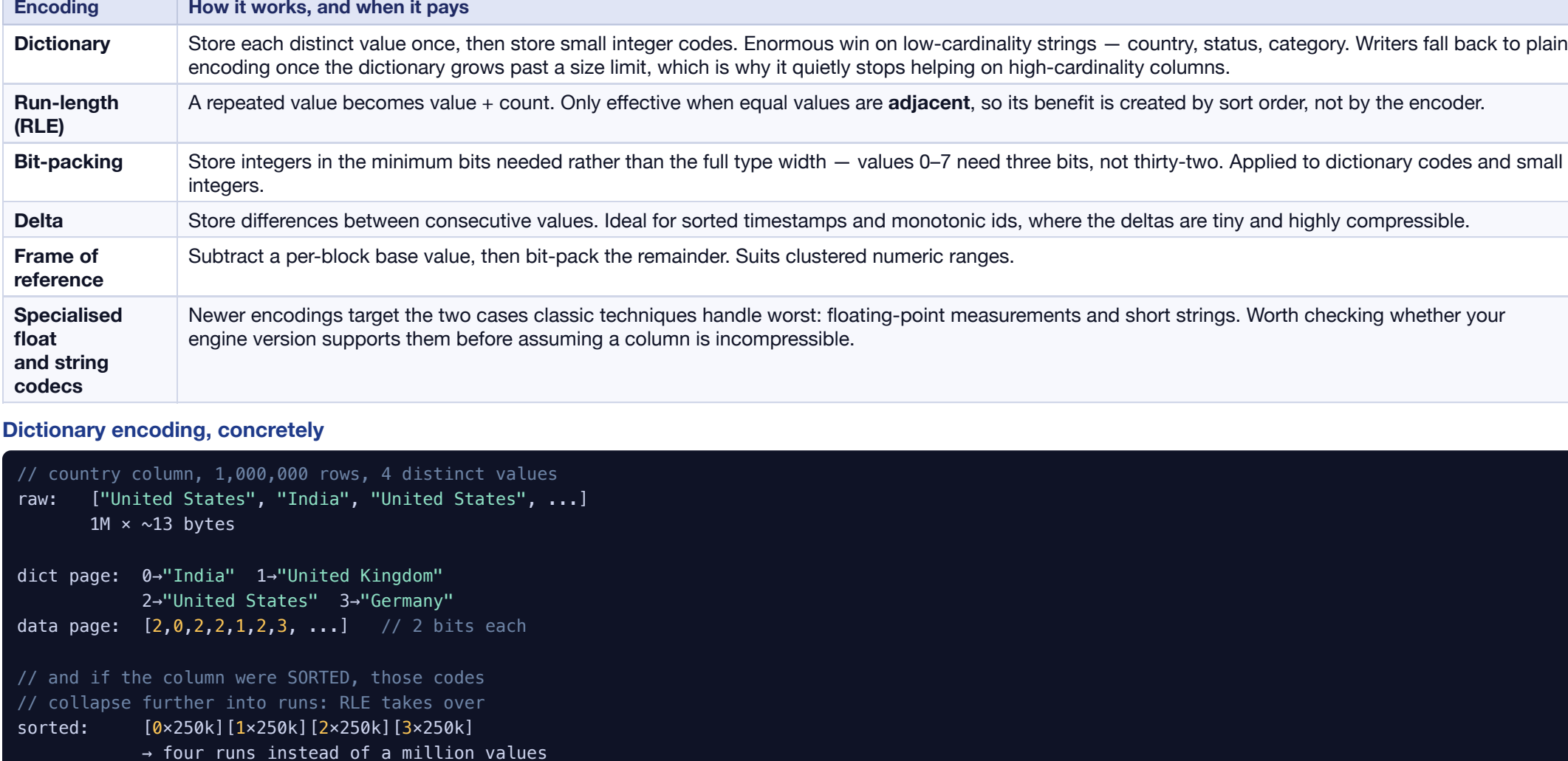
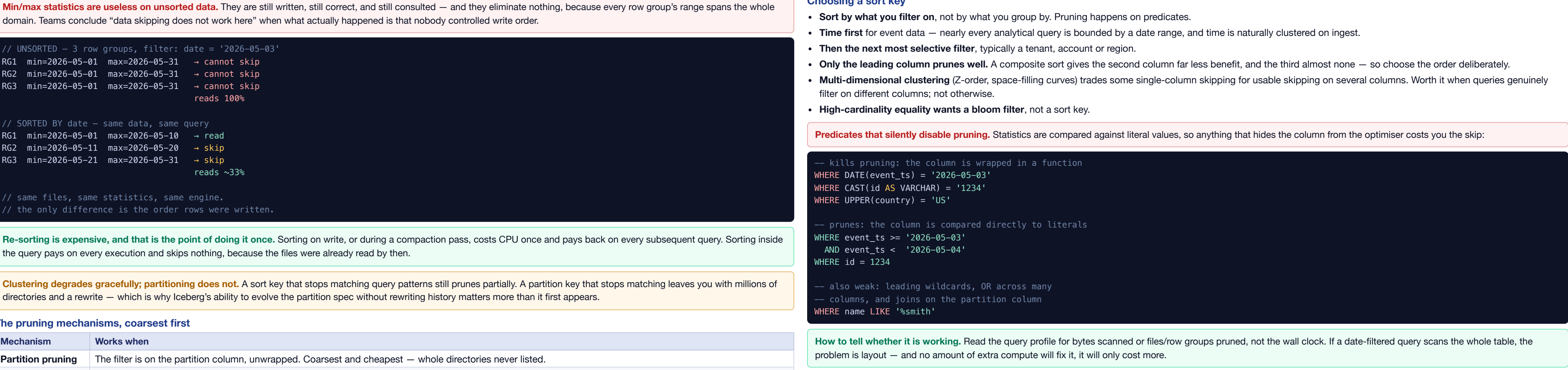
query path above · physical file anatomy in the middle · the write path below — which is the half that decides whether any of it stays fast



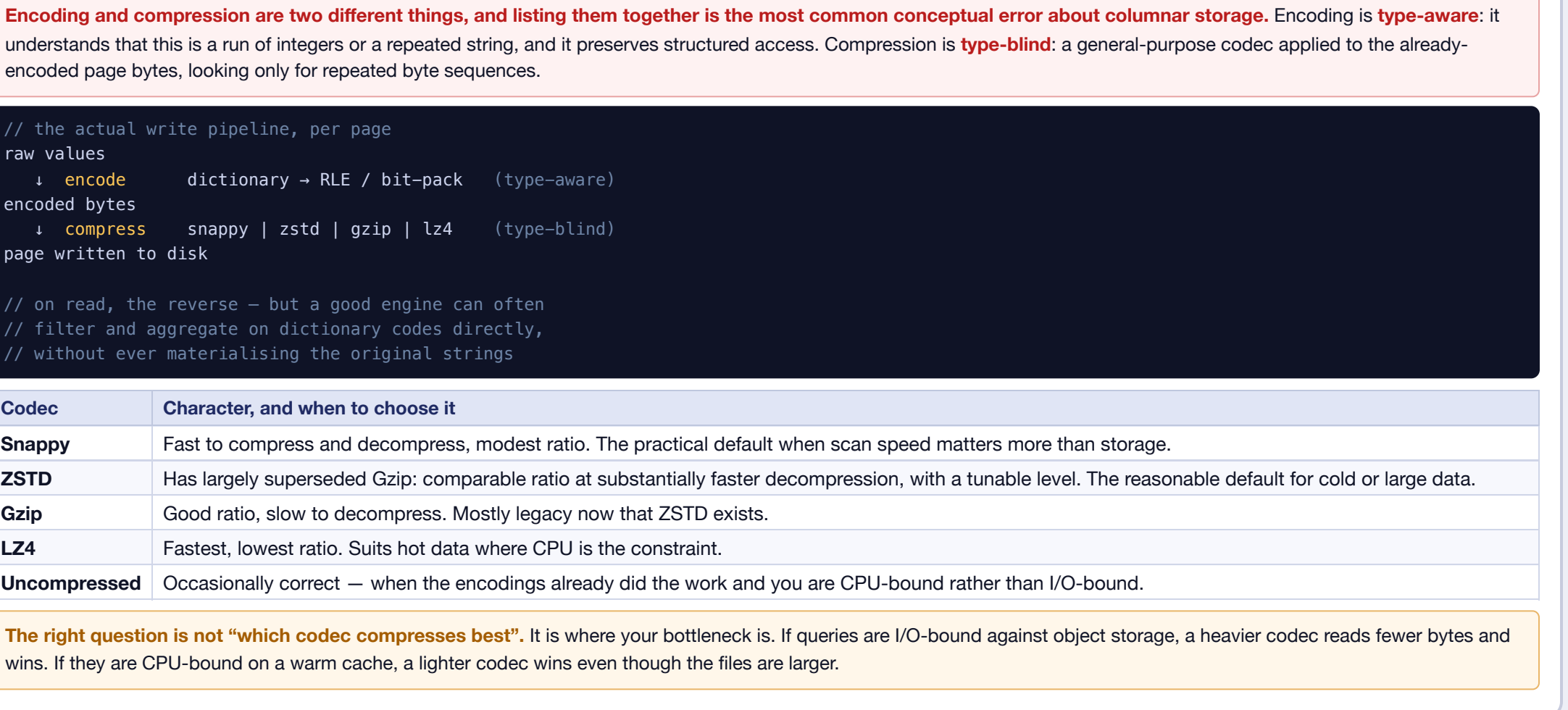
the same data, laid out differently



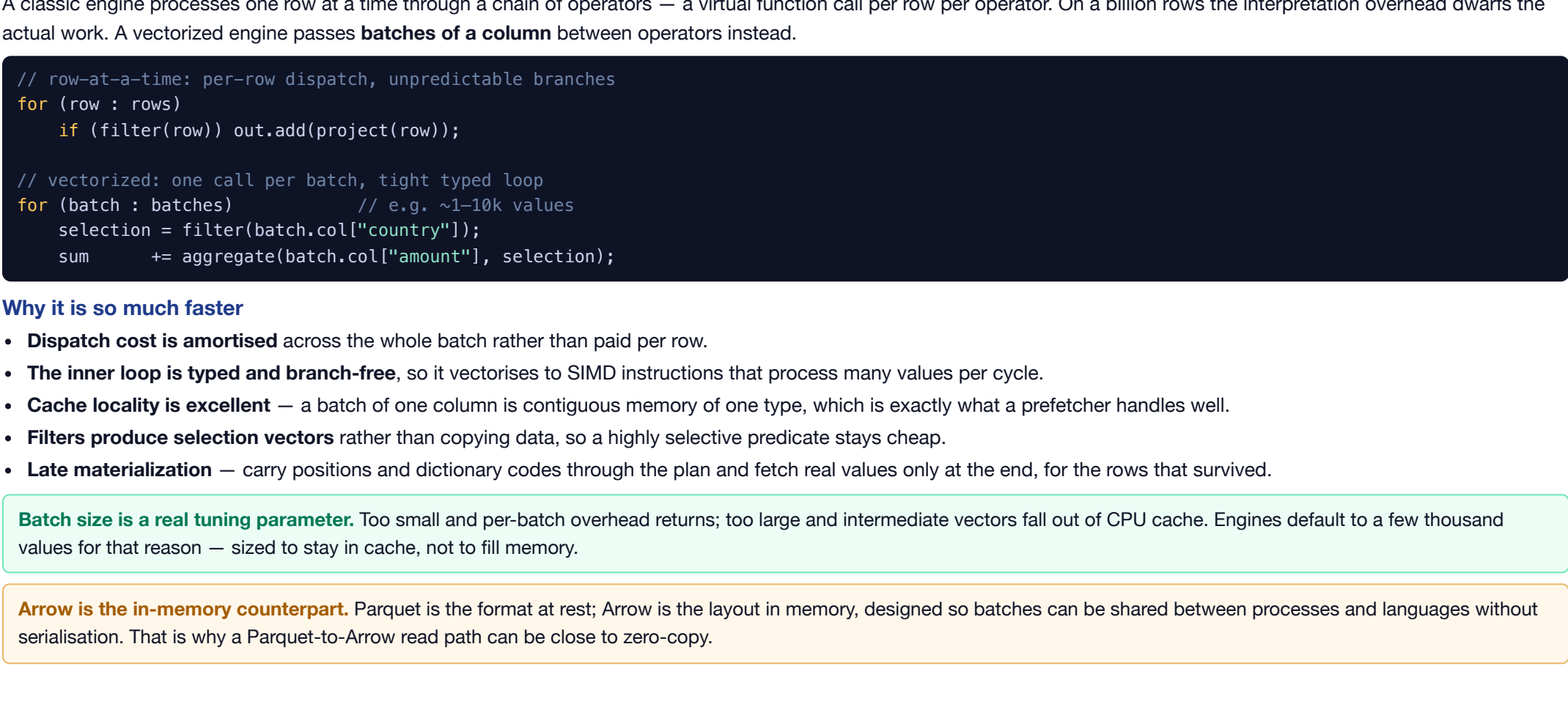
the largest single lever on columnar performance, and the one most often left untouched



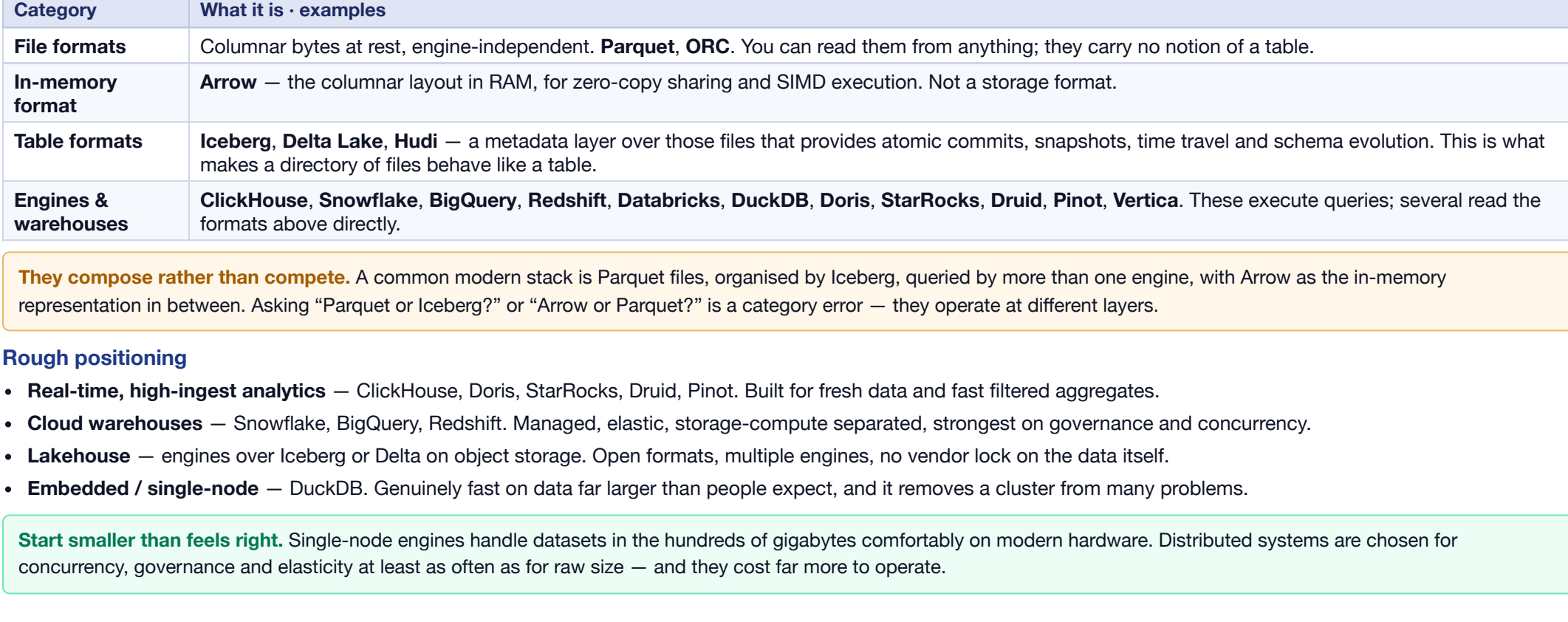
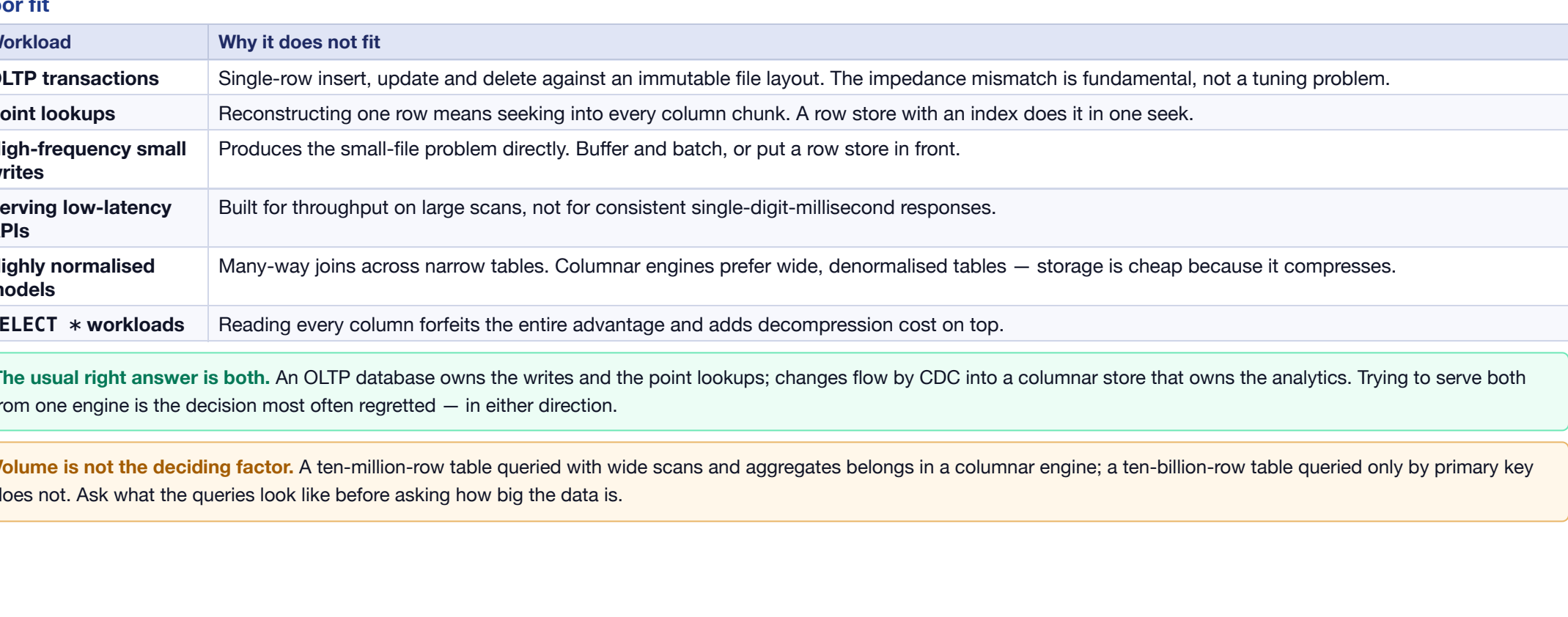
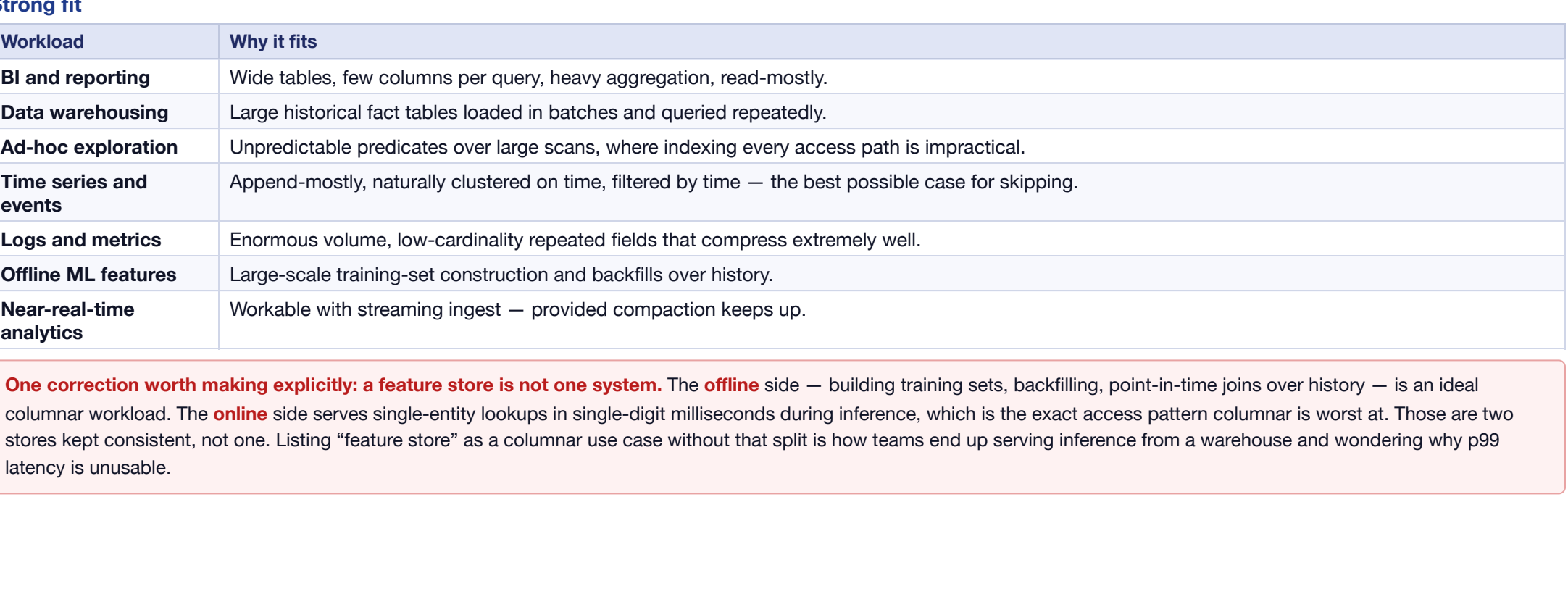
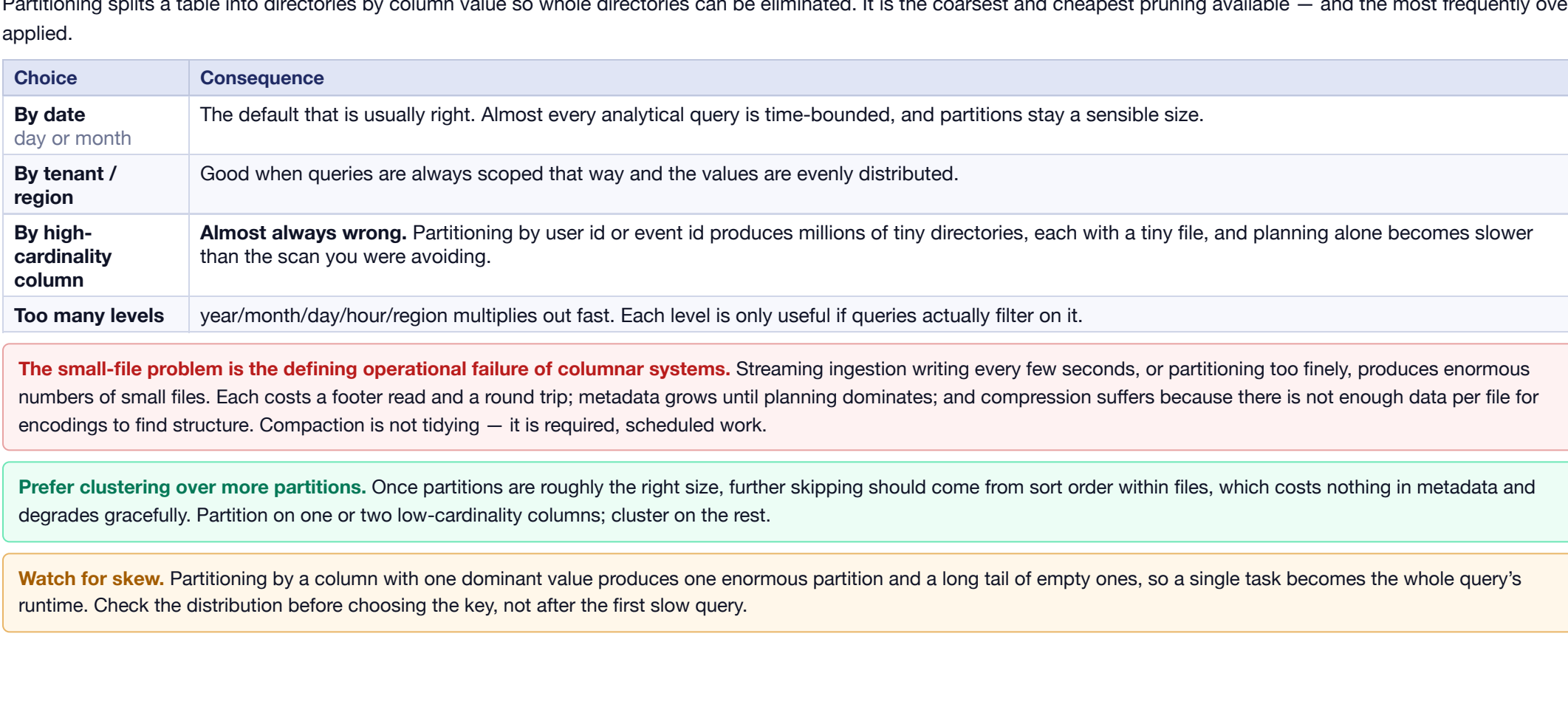
a separate stage



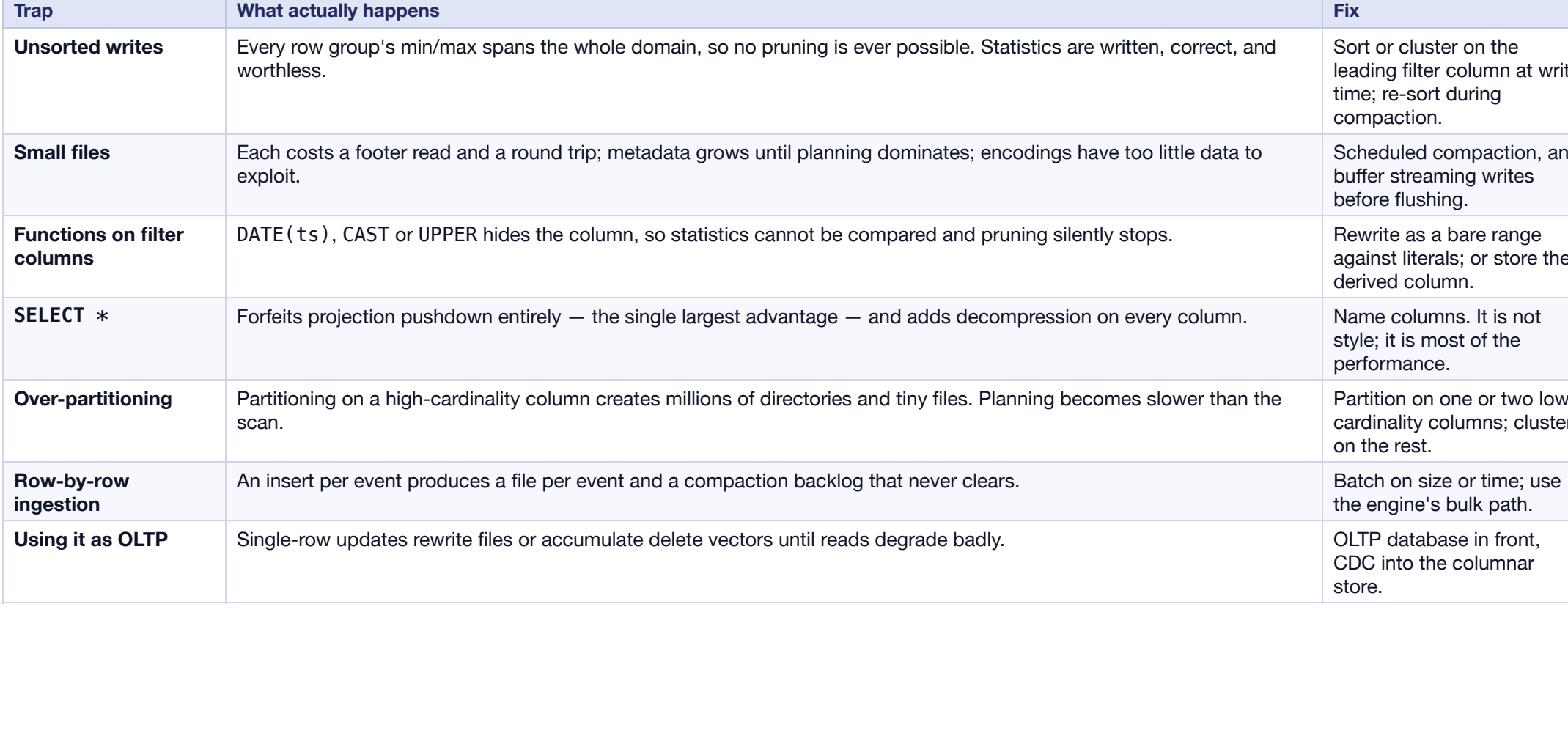
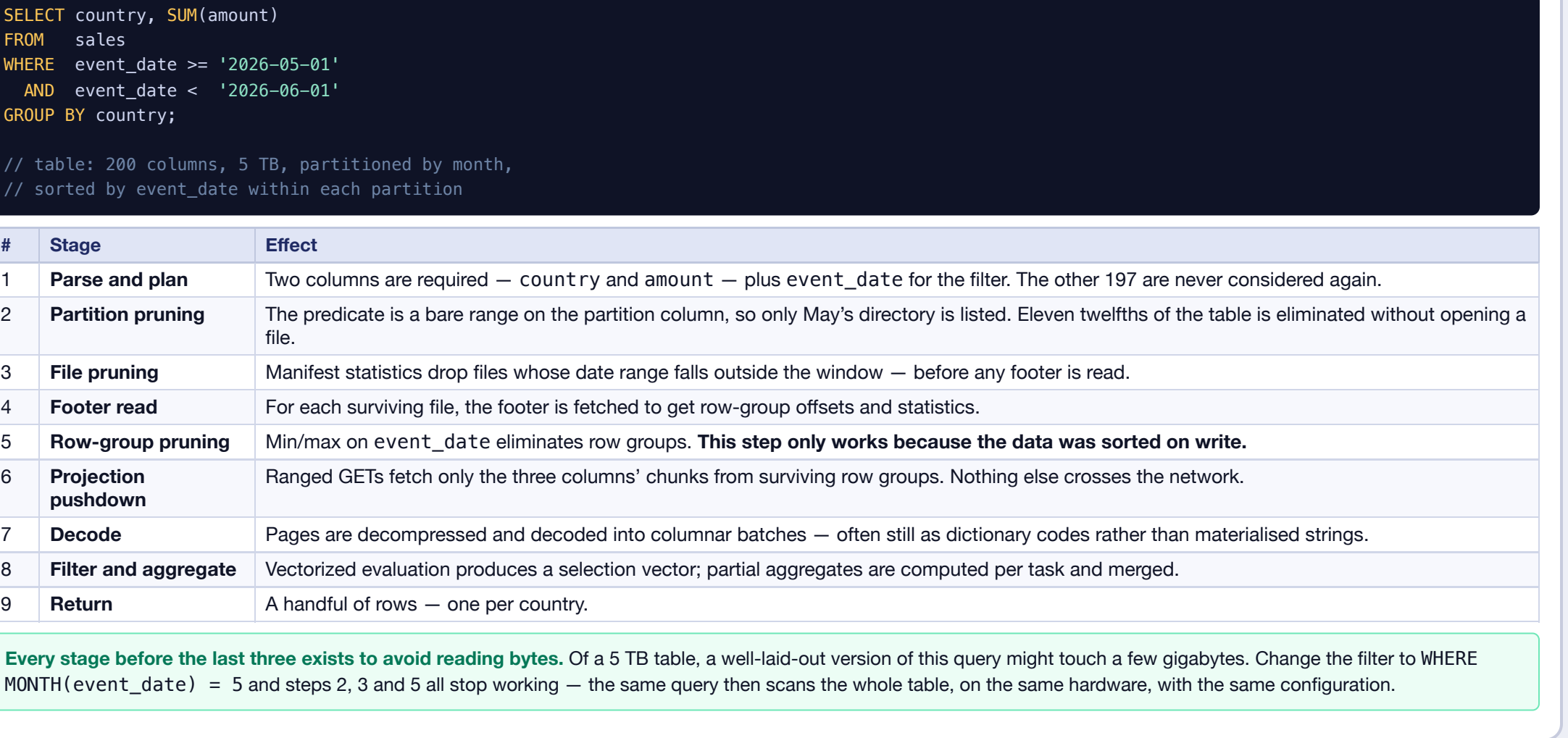
why the CPU keeps up



easy to get badly wrong



where the bytes go



confirm before a columnar table carries a workload anyone depends on

